SOC Log Analysis Investigation

Platform: LetsDefend | **Environment:** Simulated SOC log management | **Project type:** Hands-on training lab

Project Overview

I used the LetsDefend Log Management interface to locate relevant log entries, identify a source IP address associated with a requested URL, and classify a log using IP address and port information. This exercise focused on accurate log searching and interpretation rather than confirming a security incident.

Investigation 1 — Proxy Log Analysis

Task: Identify the source IP address that accessed https://github.com/apache/flink/compare.

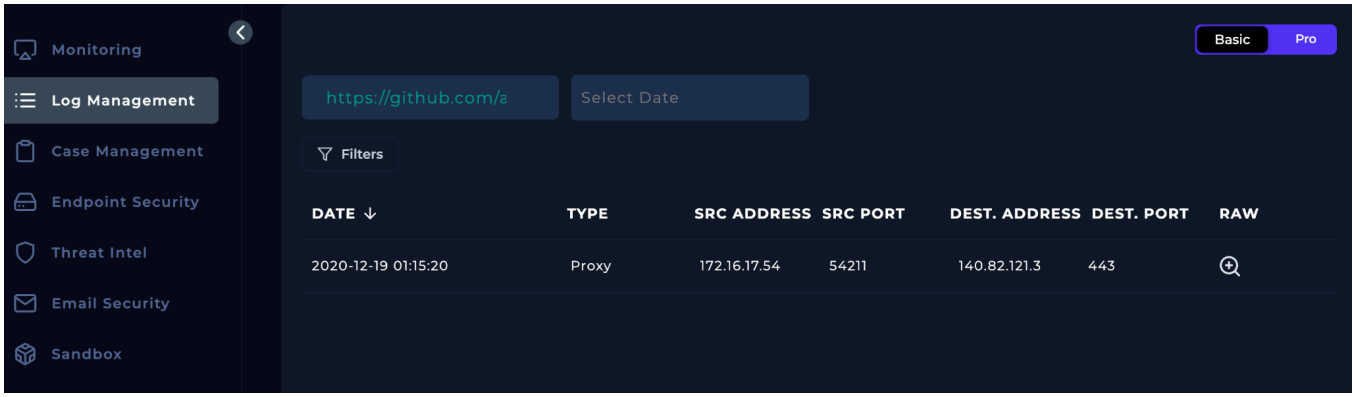


Figure 1. Filtered proxy log showing the matching network event.

Finding: The source IP address was **172.16.17.54**. The matching proxy event was recorded on 2020-12-19 at 01:15:20, with destination IP **140.82.121.3** and destination port **443**.

Investigation 1 — Result Verification

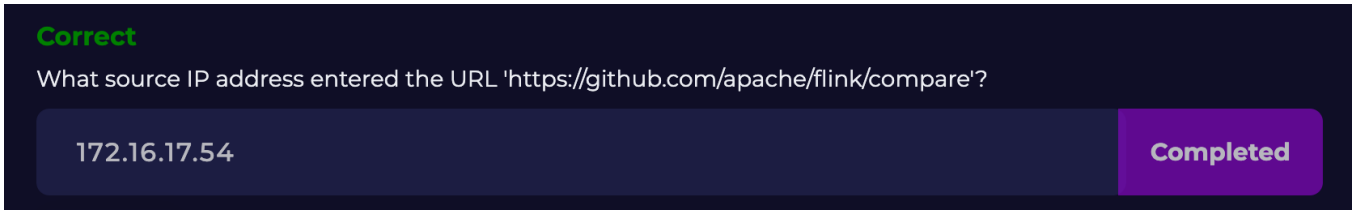


Figure 2. LetsDefend confirms the source IP answer as correct.

Investigation 2 — DNS Log Analysis

Task: Identify the log type for an event with source IP **8.8.8.8** and destination port **52567**.

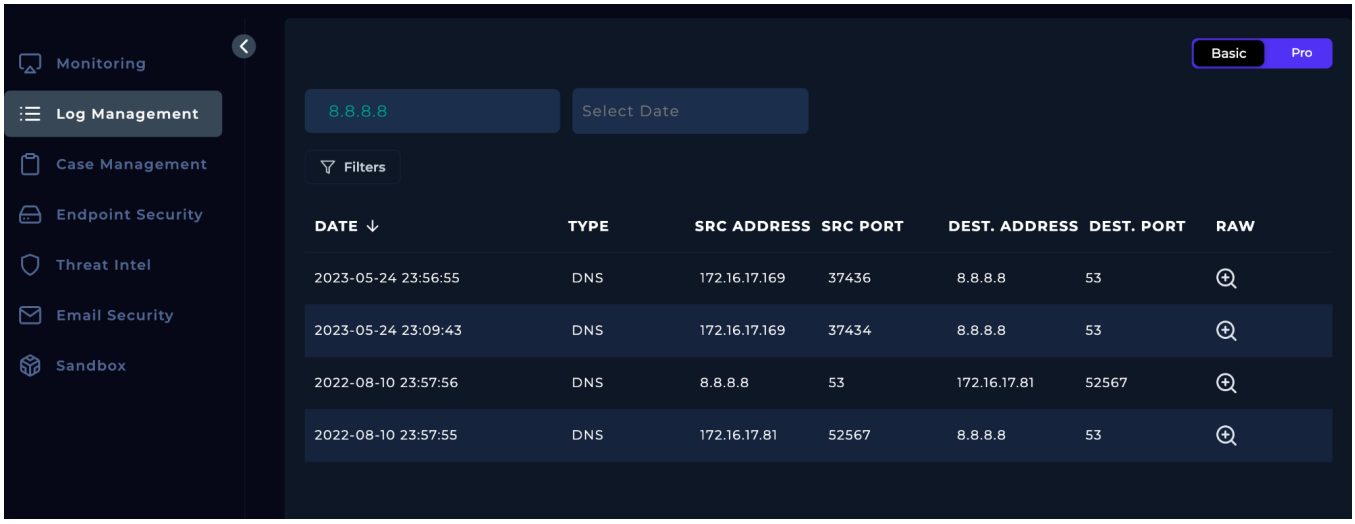


Figure 3. Search results include the matching entry with source 8.8.8.8 and destination port 52567.

Finding: The matching entry is a **DNS** log. It records traffic from **8.8.8.8:53** to **172.16.17.81:52567** on 2022-08-10 at 23:57:56. The results also show a related entry in the reverse direction one second earlier.

Investigation 2 — Result Verification

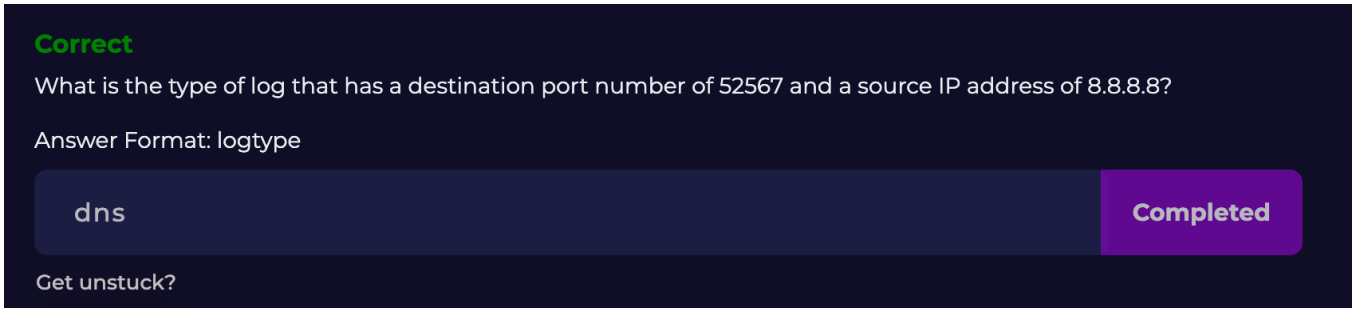


Figure 4. LetsDefend confirms DNS as the correct log type.

Skills Demonstrated

- Searching and filtering security logs by URL and IP address
- Identifying source and destination IP addresses and ports
- Distinguishing proxy logs from DNS logs
- Correlating lab questions with evidence in log records
- Documenting findings with screenshots and verification

Conclusion

Through this hands-on lab, I practiced locating specific events in security logs and extracting the relevant fields to answer investigation questions. The two tasks reinforced the importance of careful log interpretation and evidence-based reporting in SOC workflows.

Scope note: This is a simulated training exercise. The observed entries alone do not establish malicious activity, and the project is not presented as professional employment experience.